

(†) The rings should be regarded as commutative rings with identity.

Denote: $\text{Spec}A := \{ \mathfrak{p} \mid \mathfrak{p} \text{ is a prime ideal of } A \}$

We proved that

Proposition 0.1. : *There is a one-to-one correspondence between the prime ideals of $S^{-1}A$ and the prime ideals of A that is not intersecting with S :*

$$\mathfrak{p} \longleftrightarrow S^{-1}\mathfrak{p} \quad (\mathfrak{p} \cap S = \emptyset)$$

So let $f : A \rightarrow S^{-1}A$, $a \mapsto \frac{a}{1}$ be the natural homomorphism. Define

$$\psi_1 : \{ \mathfrak{p} \in \text{Spec}A \mid \mathfrak{p} \cap S = \emptyset \} \rightarrow \text{Spec}(S^{-1}A)$$

$$\mathfrak{p} \mapsto S^{-1}\mathfrak{p}$$

and

$$\psi_2 : \text{Spec}(S^{-1}A) \rightarrow \{ \mathfrak{p} \in \text{Spec}A \mid \mathfrak{p} \cap S = \emptyset \}$$

$$\mathfrak{p} \mapsto f^{-1}(\mathfrak{p})$$

then these two maps are inverse of each other.

Proposition 0.2. : *Let $\{I_1, I_2, \dots, I_n\}$ be a set of finite ideals of A , then we have*

$$(1) \quad S^{-1}(\sum_{i=1}^n I_i) = \sum_{i=1}^n S^{-1}I_i;$$

$$(2) \quad S^{-1}(\cap_{i=1}^n I_i) = \cap_{i=1}^n S^{-1}I_i;$$

$$(3) \quad S^{-1}r(I) = r(S^{-1}I)$$

Proof: (1),(2) We have proved in the case $n = 2$, by induction;

(3) For $\forall \frac{x}{s} \in S^{-1}r(I)$, i.e., $\exists n \in \mathbb{N}$ such that $x^n \in I$. Then $\frac{x^n}{s^n} = (\frac{x}{s})^n \in S^{-1}I$, i.e., $\frac{x}{s} \in r(S^{-1}I)$, hence $S^{-1}r(I) \subseteq r(S^{-1}I)$;

For $\forall \frac{x}{s} \in r(S^{-1}I)$, therefore $\exists n \in \mathbb{N}$ such that $\frac{x^n}{s^n} \in S^{-1}I$, thus $x^n \in I$ and $x \in r(I)$. So $\frac{x}{s} \in S^{-1}r(I)$ and $r(S^{-1}I) \subseteq S^{-1}r(I)$. $\square$

Proposition 0.3. : (1) $\mathfrak{n}_{S^{-1}A} = S^{-1}\mathfrak{n}_A$;

(2) let $\mathfrak{p}$ be a prime ideal of A , then there is a one-to-one correspondence between the prime ideal of $A_{\mathfrak{p}}$ and the prime ideal of A that is contained in $\mathfrak{p}$.

Proof:(1) Recall that

$$\mathfrak{n}_A = \bigcap_{\mathfrak{p} \in \text{Spec}A} \mathfrak{p}$$

then by Proposition 0.1, we get $\mathfrak{n}_{S^{-1}A} = S^{-1}\mathfrak{n}_A$;

(2) Note that $S = A \setminus \mathfrak{p}$, hence a subset K of A satisfies $K \cap S = \emptyset \Leftrightarrow K \subseteq \mathfrak{p}$. $\square$

Proposition 0.4. : *Let M be a finitely generated A -module, then*

$$S^{-1}Ann(M) = Ann(S^{-1}M)$$

Proof: By induction.

Suppose $M = \langle m \rangle$ is a cyclic module, then $Ann(M) = Ann(m)$. Then the map

$$\varphi : A \rightarrow M$$

$$a \mapsto am$$

is an A -module epimorphism and $\ker \varphi = Ann(M)$, therefore $M \cong A/Ann(M)$, thus

$$S^{-1}M \cong S^{-1}(A/Ann(M)) \cong S^{-1}A/S^{-1}Ann(M)$$

Consider

$$\psi : S^{-1}A \rightarrow S^{-1}M$$

$$\frac{a}{s} \mapsto \frac{am}{s}$$

then ψ is an epimorphism and $\ker \psi = Ann(S^{-1}M)$, hence

$$S^{-1}A/Ann(S^{-1}M) \cong S^{-1}M$$

so we get

$$0 = S^{-1}M/S^{-1}M \cong \frac{S^{-1}A/S^{-1}Ann(M)}{S^{-1}A/Ann(S^{-1}M)} = S^{-1}Ann(M)/Ann(S^{-1}M)$$

which means $S^{-1}Ann(M) = Ann(S^{-1}M)$;

For two submodules N, P of M , we have $Ann(N + P) = Ann(N) \cap Ann(P)$. Hence $S^{-1}(Ann(N + P)) = S^{-1}(Ann(N) \cap Ann(P)) = S^{-1}Ann(N) \cap S^{-1}Ann(P)$. By the proof above, we know that if N and P are both generated by one element. then $S^{-1}Ann(N) = Ann(S^{-1}N)$, $S^{-1}Ann(P) = Ann(S^{-1}P)$ and $S^{-1}(Ann(N + P)) = Ann(S^{-1}N) \cap Ann(S^{-1}P)$, note that $S^{-1}N, S^{-1}P$ are submodules of $S^{-1}M$. So we know that:

if N, P are both generated by one element, then

$$S^{-1}(Ann(N + P)) = Ann(S^{-1}N) \cap Ann(S^{-1}P) = Ann(S^{-1}N + S^{-1}P)$$

Since $M = \langle m_1, m_2, \dots, m_n \rangle$ is finitely generated, let $M_1 = \langle m_1 \rangle, M_2 = \langle m_2 \rangle, \dots, M_n = \langle m_n \rangle$, then $M = M_1 + M_2 \cdots + M_n$. $\square$

Proposition 0.5. : *If N and P are two A -submodules of M with P finitely generated, then*

$$S^{-1}(N : P) = S^{-1}N : S^{-1}P$$

Proof: We first prove that

$$N : P = Ann((N + P)/N)$$

Indeed, for $\forall a \in N : P$, $aP \subseteq N$, hence $a(N+P) = aN + aP \subseteq N$, thus $a \in \text{Ann}((N+P)/N)$ and $N : P \subseteq \text{Ann}((N+P)/N)$; for $\forall a \in \text{Ann}((N+P)/N)$, we get $a(N+P) = aN + aP \subseteq N$, since $aN \subseteq N$, therefore $aP \subseteq N$ and $a \in N : P$, $\text{Ann}((N+P)/N) \subseteq N : P$;

Note that $(N+P)/N$ is also finitely generated, so

$$S^{-1}(N : P) = S^{-1}\text{Ann}((N+P)/N) = \text{Ann}(S^{-1}((N+P)/N))$$

which means

$$S^{-1}(N : P) = \text{Ann}(S^{-1}((N+P)/N)) = \text{Ann}(S^{-1}(N+P)/S^{-1}N) = \text{Ann}((S^{-1}N + S^{-1}P)/S^{-1}N)$$

by the proof above, we know

$$\text{Ann}((S^{-1}N + S^{-1}P)/S^{-1}N) = S^{-1}N : S^{-1}P$$

so we get

$$S^{-1}(N : P) = S^{-1}N : S^{-1}P. \quad \square$$